$$t = \frac{1}{3.83\times10^{-12}} \times \ln\left[\frac{50}{37}\right]$$

$$t = \frac{0.301}{3.83} \times 10^{12} \approx 7.86 \times 10^{10} \text{ sec}$$

In years

$$t = \frac{7.86\times10^{10} s}{3.156\times10^{7} s/yr} \approx 2500 \text{ years}$$

In fact, the excavated materials were to USA sent for carbon dating by Archeological Department of Tamilnadu and the report confirmed that the age of Keezhadi artefacts lies between 2200 years to 2500 years (Sangam era- 400 BC to 200 BC). The Keezhadi excavations experimentally proved that urban civilization existed in Tamil Nadu even 2000 years ago!

9.6.7 Discovery of Neutrons

In 1930, two German physicists Bothe and Becker found that when beryllium was bombarded with α particles, highly penetrating radiation was emitted. This radiation was capable of penetrating the thick layer of lead and was unaffected by the electric and magnetic fields. Initially, it was thought as γ radiation. But in the year 1932, James Chadwick discovered that those radiations are not EM waves but they contain an charged particles of mass little greater than the mass of the proton. He called them as neutrons. The above reaction can be written as

$${}^{9}_{4}Be + {}^{4}_{2}He \rightarrow {}^{12}_{6}C + {}^{1}_{0}n$$

where ${}^{1}_{0}n$ denotes neutron.

Neutrons are fairly stable inside the nucleus. But outside the nucleus they are unstable. If the neutron comes out of the nucleus (free neutron), it decays with emission of proton, electron, and antineutrino with the half life of 13 minutes.

Neutrons are classified according to their kinetic energy as (i) slow neutrons (0 to 1000 eV) (ii) fast neutrons (0.5 MeV to 10 MeV). The neutrons with average energy of about 0.025 eV in thermal equilibrium are called thermal neutron, because at 298K, the thermal energy $kT \sim 0.025eV$. Slow and fast neutrons play a vital role in nuclear reactors.

9.7 NUCLEAR FISSION

In 1939, German scientists Otto Hahn and F. Strassman discovered that when uranium nucleus is bombarded with a slow neutron, it breaks up into two smaller nuclei of comparable masses with the release of energy. **The process of breaking up of the nucleus of a heavier atom into two smaller nuclei with the release of a large amount of energy is called nuclear fission**. The fission is accompanied by the release of neutrons. The energy that is released in the nuclear fission is of many orders of magnitude greater than the energy released in chemical reactions.

Uranium undergoes fission reaction in 90 different ways. The most common fission reactions of ${}^{235}_{92}U$ nuclei are shown here.

$${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{236}_{92}U^{*} \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3\,{}^{1}_{0}n + Q \quad (9.45)$$

$${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{236}_{92}U^{*} \rightarrow {}^{140}_{54}Xe + {}^{94}_{38}Sr + 2\,{}^{1}_{0}n + Q \quad (9.46)$$

Here Q is energy released during the fission of each uranium nucleus. When a slow neutron is absorbed by the uranium nucleus, the mass number increases by one and goes to an excited state ${}^{236}_{92}U^{*}$. But

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